Mahmudul Hasan Khan

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EDUCATION

2027	**Doctor of Philosophy, Chemical Engineering** (*Expected*) Iowa State University (ISU), Ames, IA, USA
2025	**Master of Engineering, Chemical Engineering** Iowa State University (ISU), Ames, IA, USA GPA: 3.96/4.00
2021	**Bachelor of Science, Chemical Engineering** Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh GPA: 3.62/4.00

RESEARCH EXPERIENCE

2022–Present **Graduate Research Assistant**

Department of Chemical and Biological Engineering, ISU, Ames, IA, USA

- Competitive Adsorption of Lignin-derived Oxygenates over Molybdenum Trioxide (MoO_3) Catalyst during Hydrodeoxygenation
  - Performed kinetic and mechanistic studies during the simultaneous hydrodeoxygenation (HDO) reaction of acetone and anisole over MoO_3.
  - Established a kinetic approach to quantify the relative adsorption constants of lignin-derived compounds during HDO.
  - Rationalized the role of the electron density of the reactant molecule (e.g., acetone) in dictating surface adsorption.
- Identification of Sabatier Descriptors for HDO Activity and Selectivity on Supported Metal Oxide Catalysts
  - Synthesized and characterized 15+ supported molybdenum oxide catalysts for HDO of carbonyls (i.e., ketones and aldehydes).
  - Performed activity and selectivity analysis to understand the effect of MoO_3-support material interaction on product distribution during carbonyl HDO.
  - Identified support electronegativity and surface point of zero charge as the properties that can be leveraged to modulate product selectivity.
- Understanding the Catalytic Deactivation of MoO_3 during HDO of Lignin-derived Oxygenates
  - Performed kinetic analysis of catalyst deactivation under different reaction conditions during HDO of lignin-derived model compounds.
  - Performed extensive catalyst characterization to establish a relationship between observed kinetics and structural changes in the catalyst due to the reaction.
  - Established a pretreatment protocol to synthesize a more robust HDO catalyst.
- Ambient-Pressure Conversion of Bio-derived 3-methylanisole to Benzene, Toluene, and Xylenes (BTX) over MoO_3 Catalysts
  - Screened 10+ MoO_3-based catalysts to identify the best catalyst for the efficient conversion of 3-methylanisole, an emerging byproduct of biorefineries.
  - Performed kinetic and mechanistic studies of 3-methylanisole HDO over selected best catalysts under different reaction conditions.
  - Proposed an integrated industrial setup to produce BTX aromatics from bio-derived 3-methylanisole and waste biomass.

2021 **Research Assistant**

Institute of Water and Flood Management (IWFM), BUET, Dhaka, Bangladesh

- Water Quality Monitoring (WQM) and Modeling of Transboundary Rivers in Bangladesh
  - Identified pollution trends and seasonal variations across two major transboundary river basins by assessing 100+ water and 20+ soil samples.
  - Co-authored a policy-oriented technical report and proposed a streamlined WQM framework for adoption at the national level.

2019–2021 **Undergraduate Research Assistant**

Department of Chemical Engineering, BUET, Dhaka, Bangladesh

- Ni-activated Coconut Husk Biochar as an Adsorbent for Methyl Orange Dye
  - Valorized coconut biomass using pyrolysis to Ni-doped activated carbon.
  - Performed adsorption studies to remove methyl orange dye from wastewater using the prepared activated carbon.

PUBLICATIONS

- **Khan**, M.H., Niu, M., Fretag, C., Huang, W. and Shanks, B.H., Manuscript in preparation. Investigation of the Complex Phenomenon of Molybdenum Oxide (MoO_3) Catalyst Deactivation during Anisole Hydrodeoxygenation.
- **Khan**, M.H. and Shanks, B.H., Manuscript in preparation. Selective Conversion of Bio-derived 3-methylanisole via Hydrodeoxygenation.
- **Khan**, M.H., Fard, Z.H., Kohler, A.J., Roling, L.T. and Shanks, B.H., 2025. Elucidating the Competitive Hydrodeoxygenation of Lignin-Derived Oxygenates over Bulk MoO_3 Catalyst through Kinetic Analysis. *ACS Catalysis*, 15(17), 15488–15501.
- Kohler, A.J., **Khan**, M.H. and Shanks, B.H., 2025. Identification of Sabatier Descriptors for Hydrodeoxygenation Activity and Selectivity on Supported Molybdenum Oxide Catalysts. *ACS Catalysis*, 15(5), 4317–4330.
- Sporsho, S.S., Saha, D., **Khan**, M.H., Rahman, M.S., Rukh, M., Islam, M.R., Chakma, T., Haque, F., Roy, H., Sarkar, D. and Islam, M.S., 2025. Insights into adsorbent-based pharmaceutical wastewater treatment and future developments toward sustainability. *RSC Advances*, 15(59), 50597–50632.
- **Khan**, M.H., Akash, N.M., Akter, S., Rukh, M., Nzediegwu, C. and Islam, M.S., 2023. A comprehensive review of coconut-based porous materials for wastewater treatment and CO_2 capture. *Journal of Environmental Management*, 338, 117825.

PRESENTATIONS

Oral Presentations

- **Khan**, M.H., Kohler, A.J., Fard, Z.H., Roling, L.T. and Shanks, B.H., 2024. Kinetic Analysis of Competitive Hydrodeoxygenation of Lignin-derived Molecules over Bulk MoO_3 Catalyst. American Institute of Chemical Engineers (AIChE) Annual Meeting.
- **Khan**, M.H., Kohler, A.J., Fard, Z.H., Roling, L.T. and Shanks, B.H., 2024. Kinetic Analysis of Competitive Hydrodeoxygenation of Lignin-derived Molecules over Bulk MoO_3. Great Plains Catalysis Society (GPCS) Symposium.

Poster Presentations

- **Khan**, M.H. and Shanks, B.H., 2025. Elucidating Molybdenum Trioxide (MoO_3) Deactivation during the Ambient Pressure Hydrogenation of Lignin-Derived Model Compounds. The 29th North American Catalysis Society Meeting.
- **Khan**, M.H. and Shanks, B.H., 2025. Understanding MoO_3 Catalyst Deactivation during the Hydrodeoxygenation of Lignin-derived Oxygenates. Great Plains Catalysis Society (GPCS) Symposium.
- **Khan**, M.H., Kohler, A.J., and Shanks, B.H., 2023. Kinetic Analysis of Competitive Hydrodeoxygenation of Lignin-derived Molecules over Bulk MoO_3 Catalyst. GPCS Symposium.

TEACHING AND INDUSTRY EXPERIENCE

2023 **Teaching Assistant**

Department of Chemical and Biological Engineering, ISU, Ames, IA, USA

- CH E 4060: Environmental Chemodynamics
  - Conducted weekly help sessions for a class of 30 students.
  - Formulated homework solutions and graded homework.
- CH E 4260: Chemical Engineering Laboratory II
  - Instructed and monitored mass transfer and staged operations experiments for a class of 24 students.
  - Reviewed and graded experimental reports.

2021–2022 **HSE Internship**

Chevron (Bangladesh Business Unit), Dhaka, Bangladesh

- Reviewed internal safety audits, recommended 10+ corrective actions, and contributed to improved compliance with Chevron's safety standards, helping reduce flagged items in the annual review.
- Updated the regulatory compliance workbook to reflect the latest government policies and developed concise safety briefing materials used in monthly "Safety Moment" meetings, enhancing workforce engagement on operational risk awareness.

TECHNICAL SKILLS

- **Catalysis and Reaction Engineering:** Reaction kinetics and mechanism development; packed-bed reactor (PBR) design and operation; catalyst deactivation diagnosis and stabilization strategies; transport analysis and verification of kinetic regime (internal and external mass transfer limitations); structure–activity–selectivity relationships; incipient wetness impregnation; wet impregnation; supported metal oxide catalyst synthesis and screening.
- **Characterization and Analysis:** Gas chromatography (GC, GC-MS); chemisorption (TPR, TPD, pulse titration); BET physisorption; X-ray diffraction (XRD); X-ray photoelectron spectroscopy (XPS); Raman spectroscopy; FTIR; TEM; SEM; ICP-OES.
- **Data and Statistics:** MATLAB; Python (NumPy, SciPy, Pandas); JMP; R; MS Excel; OriginPro; Power BI; CasaXPS; Design of Experiments (DoE: full/fractional factorial); Statistical Process Control (SPC); root cause analysis.

AWARDS, GRANT PROPOSALS, AND FELLOWSHIPS

2026 **CBE Perfect Pitch Award**

Awarded second place by the Department of Chemical and Biological Engineering at ISU after a competitive 3-minute thesis-style presentation.

2025–2026 **Sun Grant Program (SGP)-North Central Region (USDA–NIFA)**

Submitted the proposal "Investigation of the Complex Phenomenon of Molybdenum Oxide (MoO_3) Catalyst Deactivation during Ambient Pressure Hydrodeoxygenation" as lead proposal author, with Prof. Brent Shanks as PI. Status: Not selected for funding.

2022 **George W. Parrott Centennial Graduate Fellowship, ISU**

The fellowship was awarded to one of 80 applicants, selected based on their undergraduate academic records and research experience.

2022 **CBE Excellence Fellowship**

The fellowship was awarded to graduate students based on their academic performance by the Department of Chemical and Biological Engineering at ISU.

SERVICE AND PROFESSIONAL MEMBERSHIPS

- **Leadership:** Vice President (2024–25) and Professional Development Chair (2023–24), Chemical Engineering Graduate Student Organization, ISU.
- **Memberships:** NSF Engineering Research Center for Biorenewable Chemicals (CBiRC), ISU; American Institute of Chemical Engineers (AIChE); Great Plains Catalysis Society (GPCS); Tau Beta Pi, Iowa Alpha.